Technical implementation & system reliability

How it's built



Listen: Kafka prices + news; waits for all 40 prices of the day



Score: EMA, RSI, ATR, volume flow, news → −1 to +1



Risk guard: 8% / 15% drawdown, 3% daily loss, STOP / FLATTEN



Plan: ≤ 10 orders, each checked against cash and platform rules



Execute: parallel orders; books synced with the server

When things fail

Kafka stalls → reconnects with backoff; watchdog rejoins after 20 s

Order times out → looked up on the server, never resent blind

3 rejects in a row → orders paused 60 s, books resynced

Too many requests → pacer: 12 per 5 s; heartbeats go first

Crash or restart → positions, cash and tax rebuilt from the server

Bad stock data → screened out before any money goes in

4,772 requests, 0 errors

0 unknown orders

₹0 gap vs server wealth

96 · 21/21 tests · checks

Strategy innovation & analytical depth

What's new

Volume-flow core

Best 10 of 40 by buying vs selling volume; more money where buyers lead.

Smart entry

Buy only while buyers lead; anything still waiting is bought by day 20.

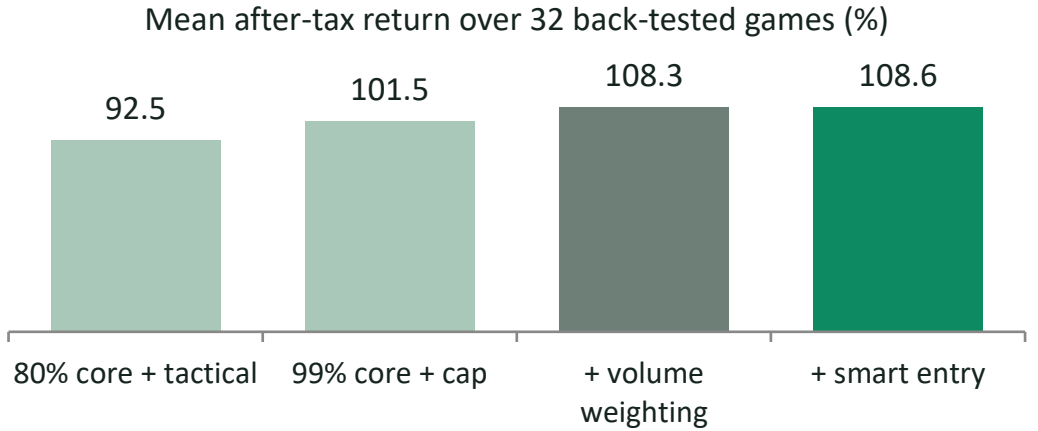
Tax as strategy

Tax-free gain harvest, loss harvest and long-term holding in every plan.

Sell-off hedge

Shorts only when ≤ 30% of stocks rise, outside the core, ≤ 5 days.

How we know it works



32 games, not one lucky path: a one-day shift moved one game from ₹12.2 L to ₹14.4 L

Train / test split: 9 popular ideas rejected; news had no next-day edge (t = 0.56)

Unseen market: 40 new stocks in pre-flight, +107.9% vs the market's +81.5%

Market friction & tax optimization

Keeping costs low

Cost	Our control
₹20 per order	Orders ≥ ₹25,000, so fees stay < 0.1%
₹15.34 per sale	Core held through dips: few sales
100% short margin	Shorts only in sell-offs, ≤ 5 days
10 orders a day, no cancel	Checked before sending; sells first
Idle cash	Only 0.5% kept back for fees

₹3,613
all charges over 1,008 days: 0.17% of final wealth

₹299
gained for every ₹1 of fees

Paying less tax, legally

1. **Hold > 1 year:** 10% tax instead of 15%
2. **Wait for the 1-year mark:** delay sales within 30 days of it
3. **Tax-free gain harvest:** use the ₹1 L allowance, re-buy at once
4. **Tax-loss harvest:** book losses against gains, re-buy same day
5. **Cap trims:** sell long-term holdings first
6. **Restart-safe:** tax history reloaded from the server

₹1.97 L
profit booked at ₹0 tax (replay)

₹21.7k
tax saved by harvesting ₹1.60 L of losses

0.9%
of final wealth paid as tax (est. ₹18,673)

Execution speed & capital efficiency

Speed

~40 ms
to compute a full day's decision

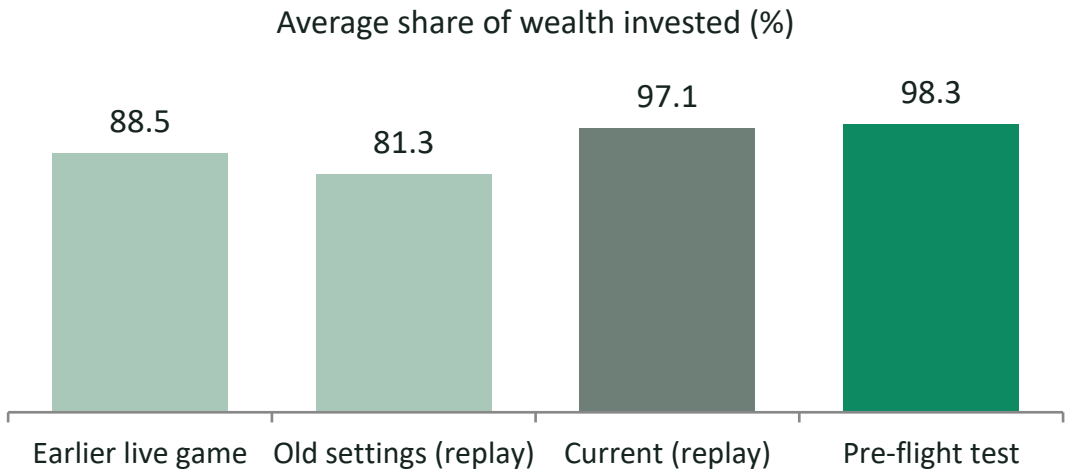
424 ms
median, first price to decision

101 ms
median order round trip

< 0.6 s
95% of days; a day lasts ~4 s

Most of the time is a deliberate wait for all 40 prices; orders go out in parallel, and every order is checked locally first.

Capital efficiency



99.5%
invested by day 3

0.76
Sharpe; market 0.68

41%
max fall; market 35%